7.1.6 Software Integration Process

NOTE — The Software Integration Process in this International Standard is a lower-level process of the Software Implementation Process.
Users of ISO/IEC 15288 may decide that this process is to be provided by the Integration Process of ISO/IEC 15288 in a recursive application of that standard.

7.1.6.1 Purpose

The purpose of the Software Integration Process is to **combine software units and software components** into integrated software items, consistent with the software design, and to demonstrate that the **functional and non-functional software requirements** are satisfied on an equivalent or complete operational platform.

7.1.6.2 Outcomes

As a result of successful implementation of the Software Integration Process:

a) an integration strategy is developed for software units consistent with the software design and prioritized software requirements;
b) verification criteria for software items are developed to ensure compliance with the software requirements allocated to the items;
c) software items are verified using the defined criteria;
d) software items defined by the integration strategy are produced;
e) results of integration testing are recorded;
f) consistency and traceability are established between the software design and the integrated software items; and
g) a regression strategy is developed and applied for re-verifying software items when changes to software units (including associated requirements, design, and code) occur.

7.1.6.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Software Integration Process.

7.1.6.3.1 Software Integration

For each software item (or configuration item, if identified), the software integration activity consists of the following tasks:

7.1.6.3.1.1 Integration Planning

The implementer shall develop and document an **integration plan** describing:

- the integration strategy and sequence;
- roles and responsibilities;
- integration schedule;
- required resources, tools, and environments;
- test requirements, procedures, and data;
- acceptance criteria for each integration increment; and
- regression testing strategy.

The integration plan shall be approved and placed under configuration management.

7.1.6.3.1.2 Integration Execution

The implementer shall integrate the software units and software components in accordance with the integration plan. Testing shall be performed incrementally as aggregates are built to ensure each aggregate satisfies its allocated requirements. Upon completion of the activity, the integrated software item shall be fully assembled. Integration and test results shall be documented.

NOTE — Integration test reports, logs, and evidence should be stored in /docs/audit/integration/ or an equivalent controlled repository.

7.1.6.3.1.3 Documentation Update

The implementer shall update the user documentation as necessary to reflect the integrated software item.

7.1.6.3.1.4 Regression Testing

The implementer shall **develop and apply a regression testing strategy** to re-verify the software items when changes are made to software units, requirements, design, or code. Regression test results shall be recorded and archived.

7.1.6.3.1.5 Evaluation of Integration Artifacts
The implementer shall evaluate the integration plan, design, code, integration test procedures and reports, test results, and user documentation considering the criteria listed below. The results of the evaluations shall be documented:

a) traceability to the system and software requirements;
b) external consistency with the system and software requirements;
c) internal consistency of the integrated software item;
d) test coverage of the software item's requirements;
e) appropriateness of test standards and methods used;
f) conformance of test results to expected outcomes;
g) feasibility of subsequent software qualification testing;
h) feasibility of operation and maintenance.

7.1.6.3.1.6 Reviews
The implementer shall conduct reviews in accordance with subclause 7.2.6.

Examples of relevant reviews include integration readiness reviews, design reviews, and regression strategy reviews.

7.1.6.4 Deliverables

The following deliverables shall be produced during the Software Integration Process:

- Approved Integration Plan
- Integrated software item(s) and configuration baselines
- Integration test procedures, data, and results
- Regression test strategy and results
- Updated user documentation
- Integration evaluation reports
- Integration review records (e.g., readiness review minutes)
- Integration Audit Log (IAL)

7.1.6.5 Audit and Configuration Control

All integration artifacts (integration plan, procedures, results, review records, regression strategy, etc.) shall be placed under configuration management. Audit evidence shall be retained for at least one release cycle or longer if required by organizational or regulatory policy. Digital signatures or equivalent measures shall be used to ensure evidence authenticity and integrity. Updates to configuration baselines shall be documented after each integration increment.